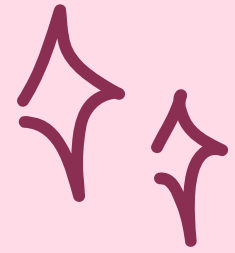


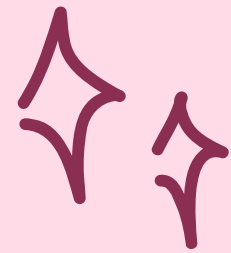
**RUBE
GOLDBERG
MACHINE!**



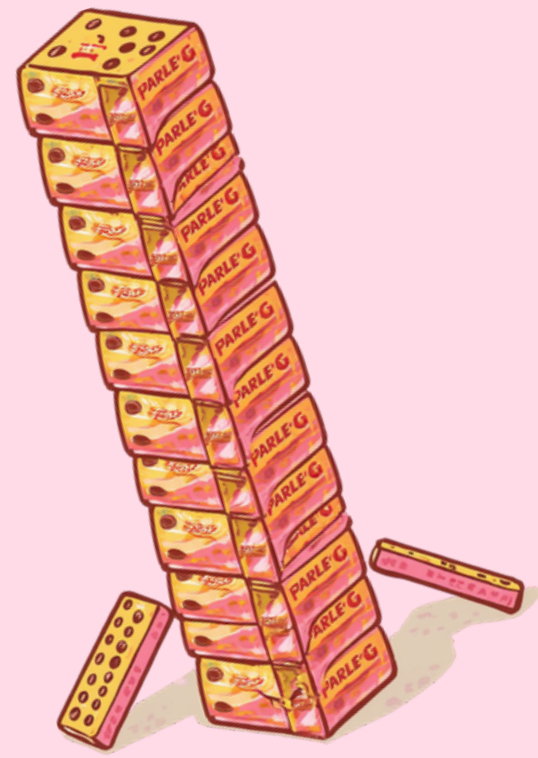
WONDER QUESTION

Can you ring a bell **WITHOUT** using your hands?
Think about it! Is there a way to make a bell go **DING**
without touching it? What could you use instead? Let your
imagination run wild!

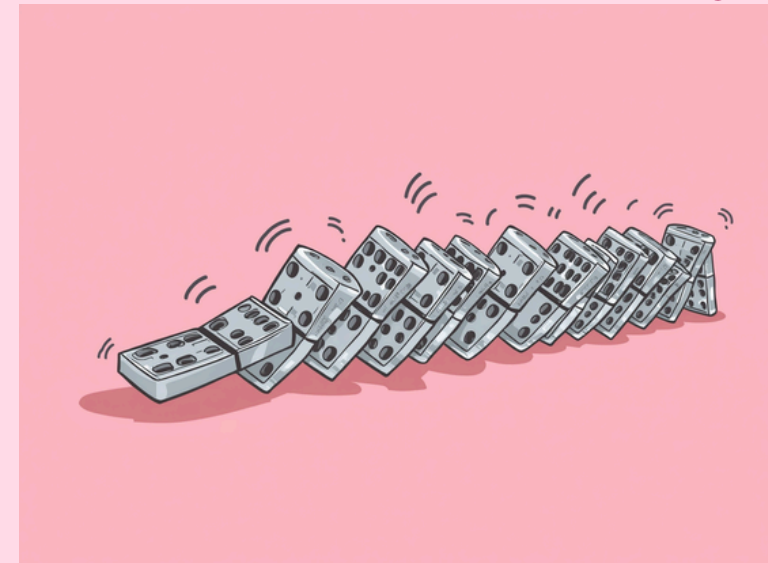
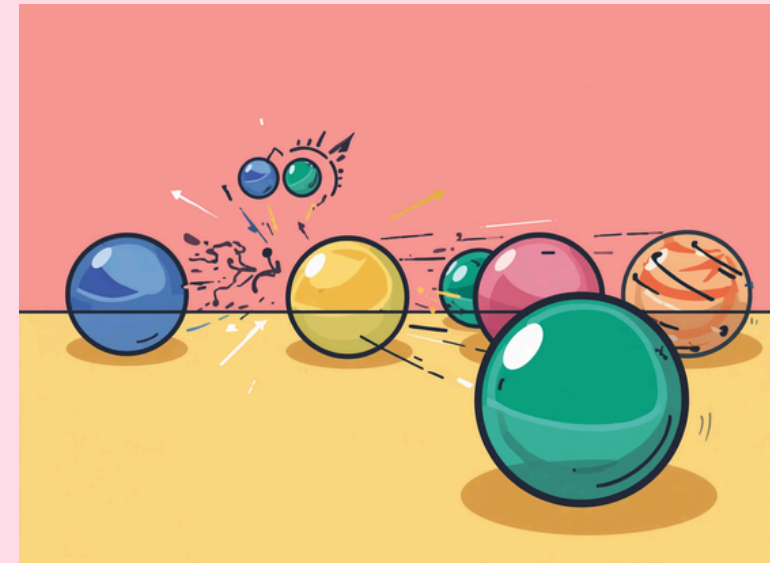




What is a Chain Reaction?



A chain reaction is when one thing causes another,
then another, then another!
Imagine steel tumblers falling in a row -
CLANG! CLANG! CLANG!
Or Parle-G biscuit packets lined up like dominoes
- one falls and they ALL fall!
Or a marble rolling and hitting other marbles -
they all start moving!





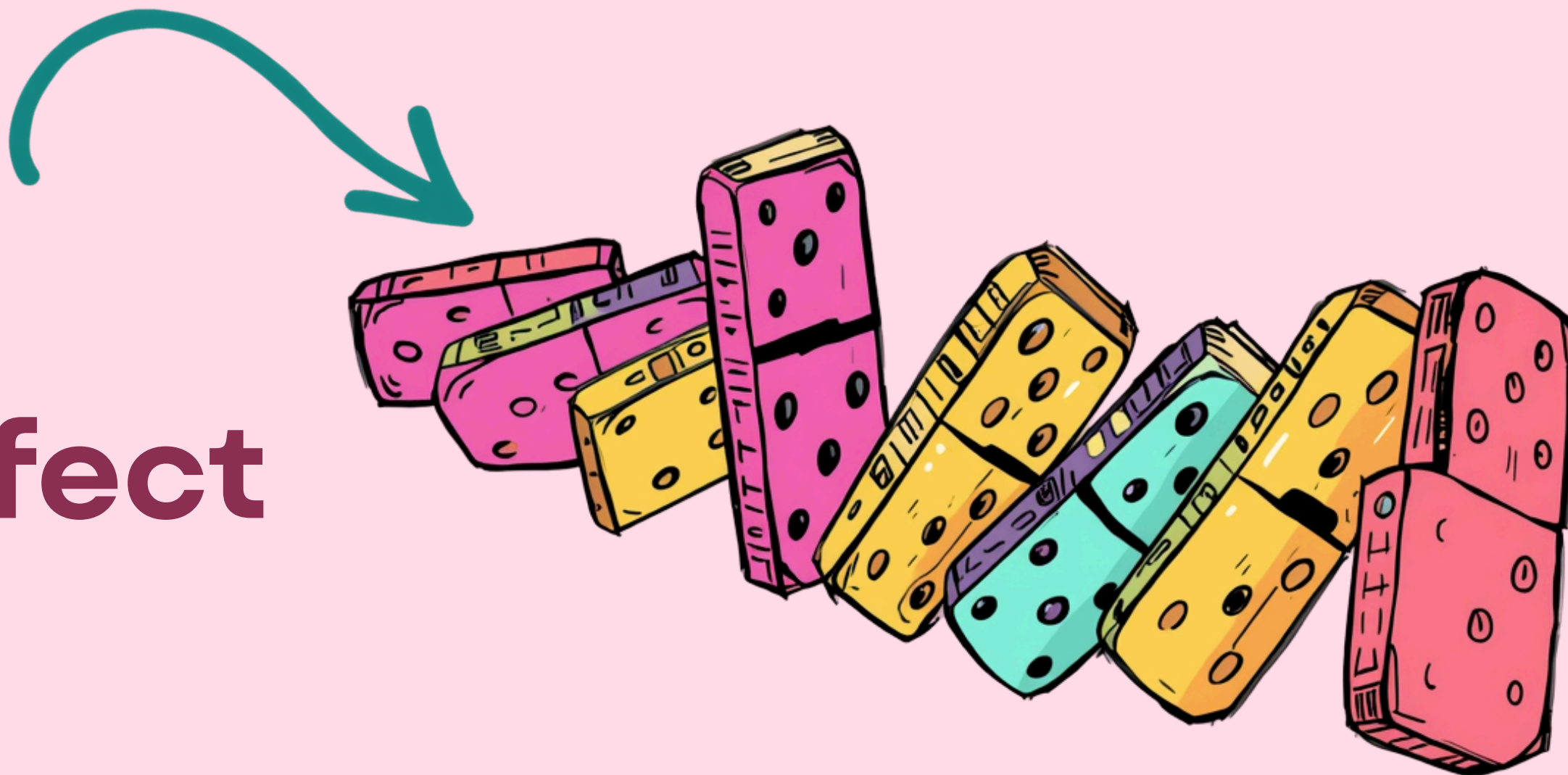
Cause and Effect

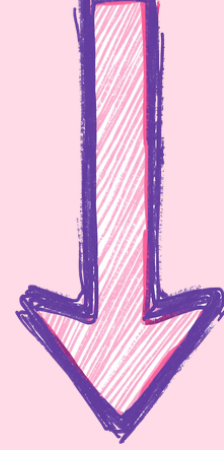
When I push THIS... THAT happens!

Push a glass → It falls down!

Knock one domino → The next one falls too!

⚽ Roll a ball down a ramp → It hits something at the bottom! See? One action causes another action to happen!





Arjun's Biscuit Story



Arjun knocked over his Parle-G packet → Hit the water bottle → Bottle rolled → Knocked the pencil box → Lid popped open → Pencil rolled out → Poked his sister!
ONE accident caused FOUR things to happen! That's a chain reaction!





FIRST → THEN → NEXT → FINALLY

The sequence matters!

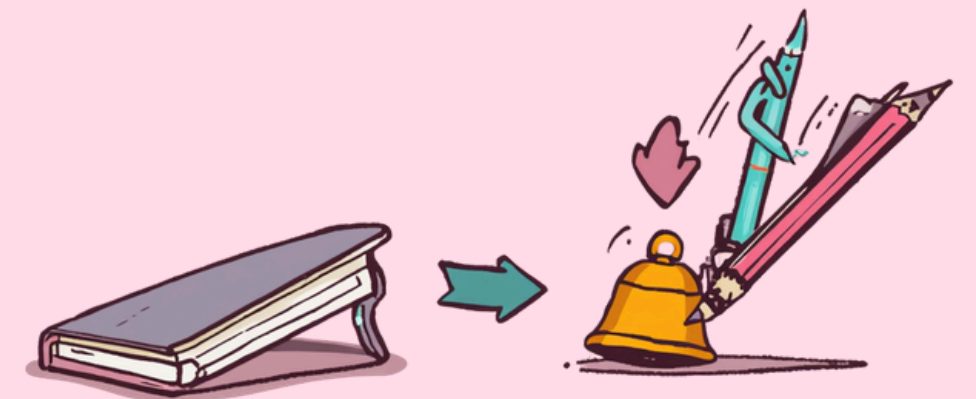
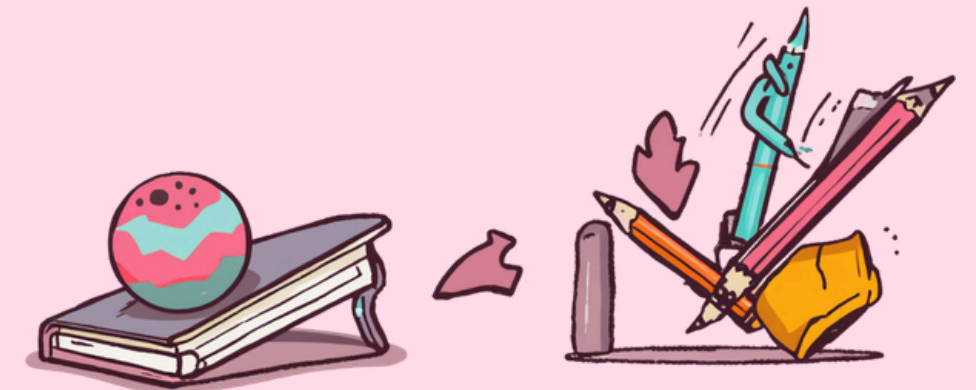
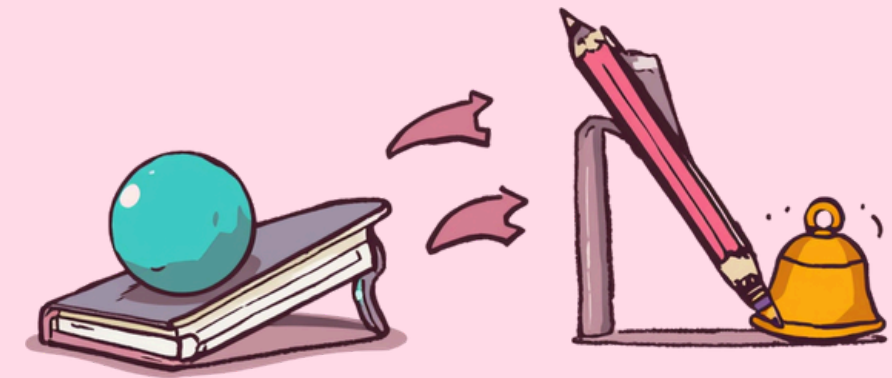
FIRST: Marble sits at the top of a tilted book

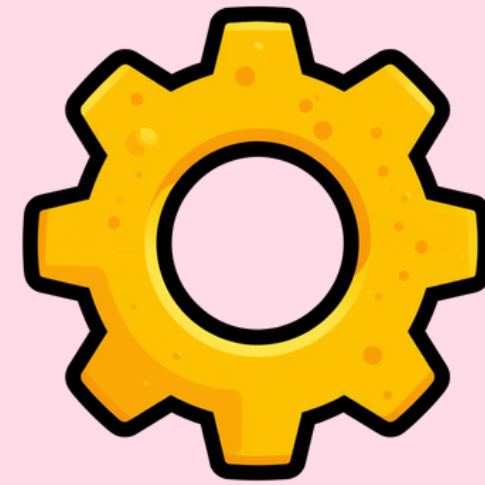
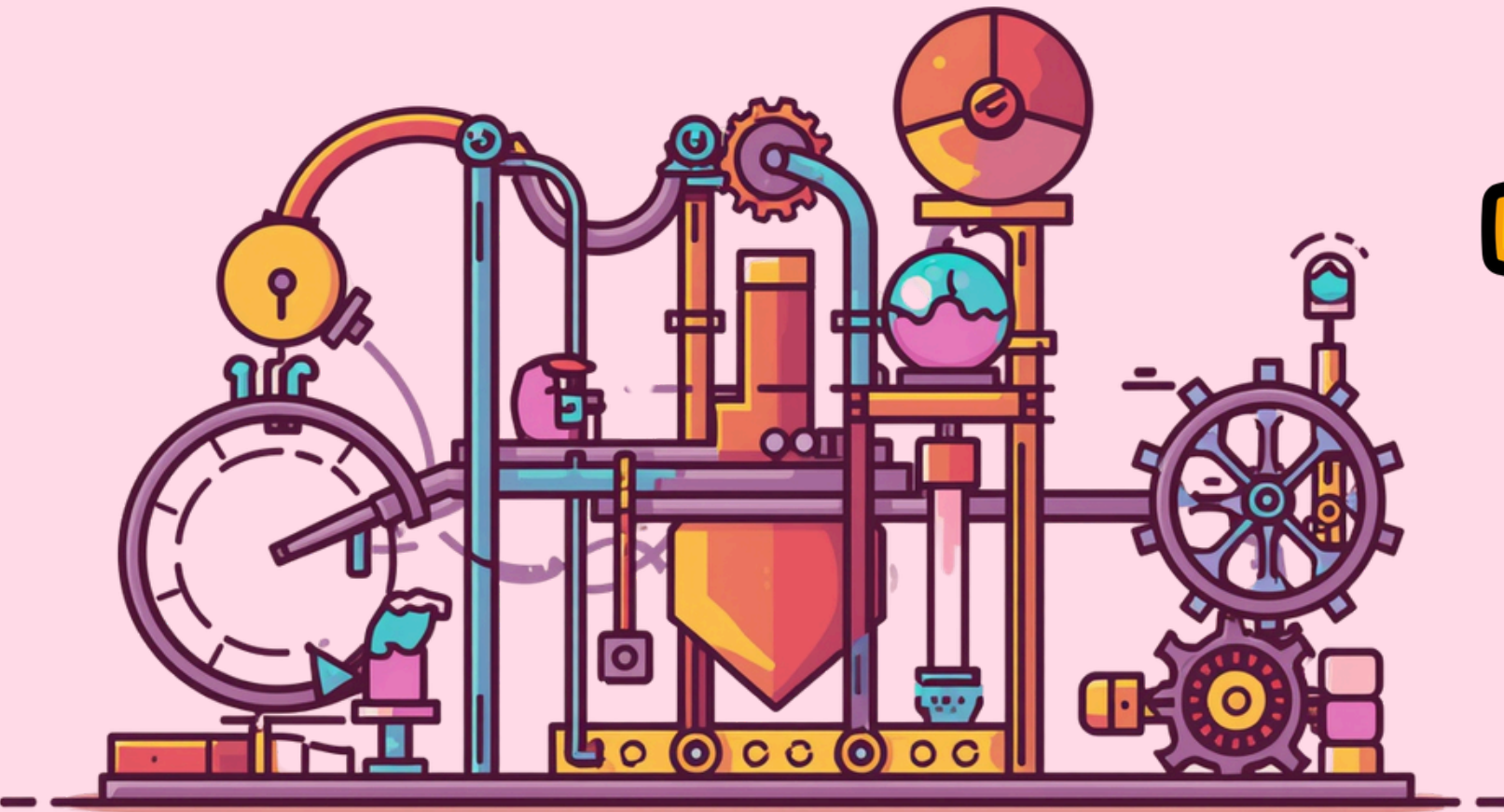
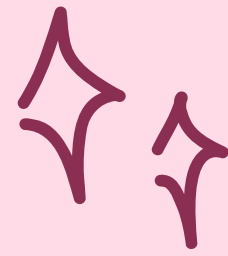
THEN: Marble rolls down the ramp

NEXT: Marble hits a row of standing pencils

FINALLY: Pencils knock the bell - DING!

Each step must happen in the right order for the chain reaction to work!



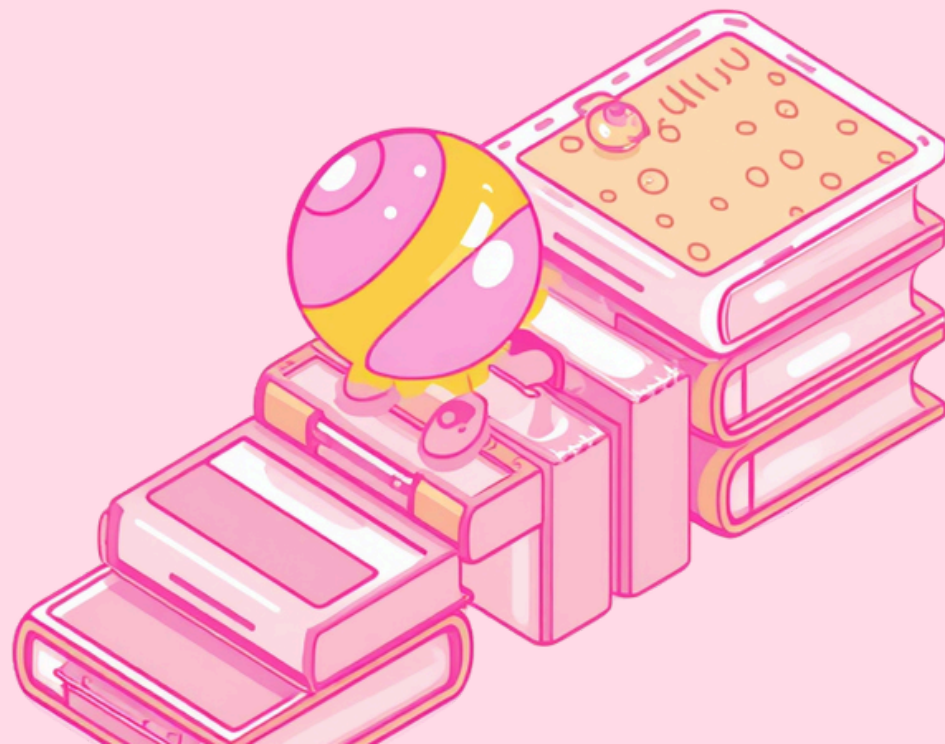


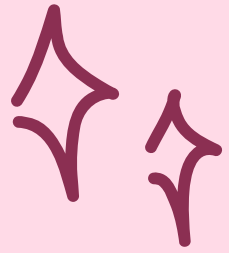
What is a Rube Goldberg Machine?

A Rube Goldberg Machine is a chain reaction machine that does a simple job in a funny, long way!

It's like a story with many steps!

Imagine using 10 things just to ring a tiny bell - that's the fun part!





Simple Tasks Our Machine Can Do

- Ring a small ghanti (temple bell)
- Drop a coin into a gullak (piggy bank)
- Push a small toy car
- Knock over a cup
- Move a small flag

The whole big machine... just to do THIS tiny thing?
That's the POINT!



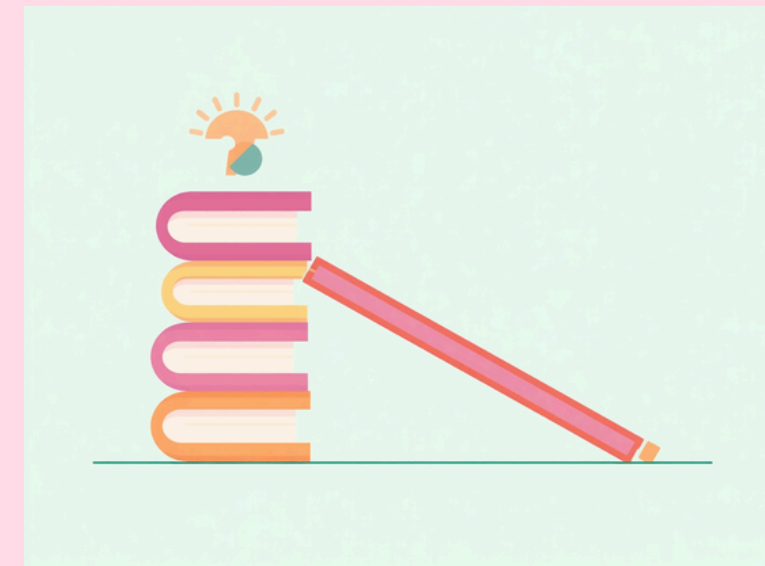
Let's Build!

What you need:

- 2-3 books (for ramps)
- 1 small ball or marble
- 3-4 blocks or steel tumblers
- 1 small toy (the grand finale!)

Steps:

- Make a ramp with books
- Line up blocks in a row
- Place your special object at the end





Test Your Machine!



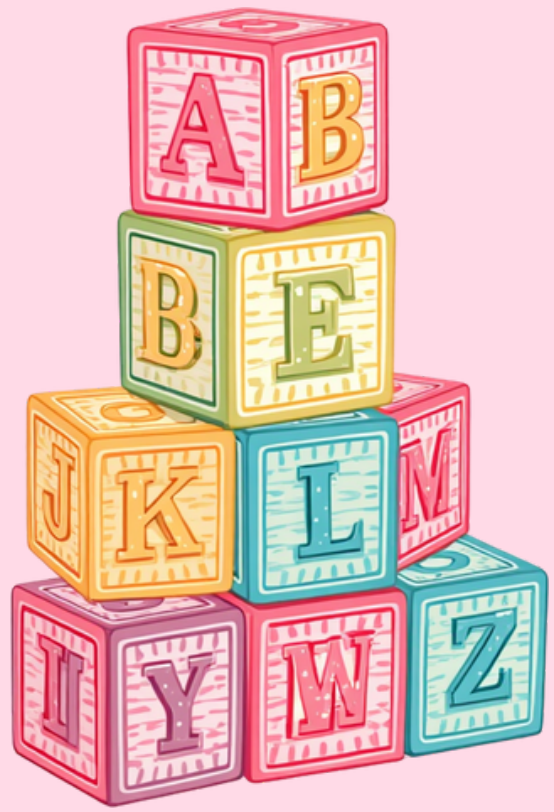
Ready? 3... 2... 1... RELEASE THE BALL!

Did it work? AMAZING!

Didn't work? EVEN MORE AMAZING! Now you're a REAL engineer! Let's fix it and try again!

Every time you test and try again, you learn something new. That's what real inventors do!





Vocabulary Time!

CHAIN REACTION

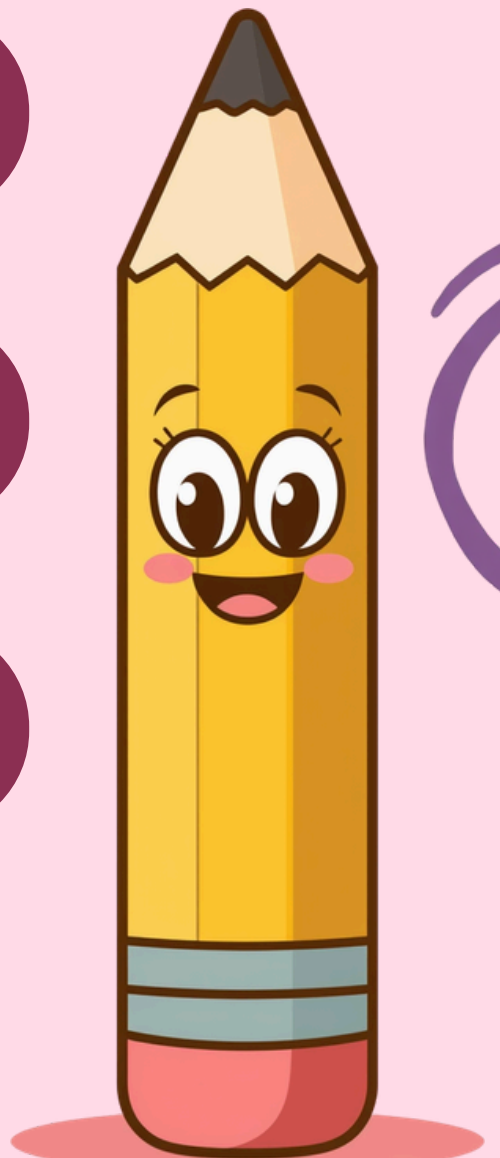
CAUSE

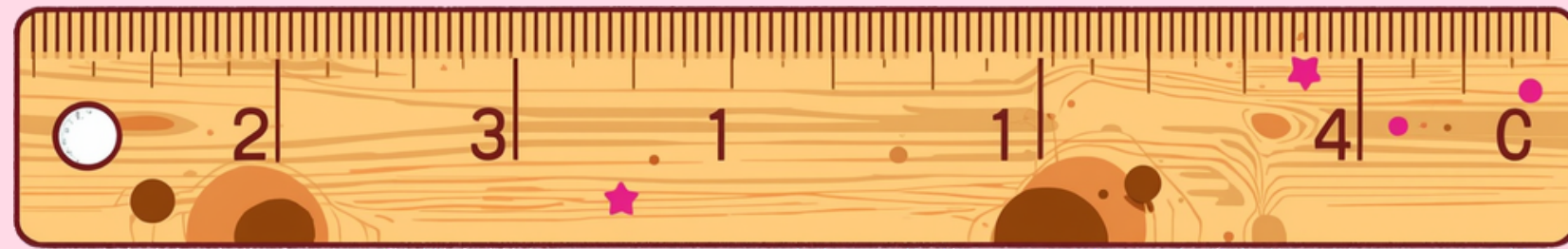
One thing causes another!

EFFECT

Fun complicated machine!

RUBE GOLDBERG





You Are An Inventor!



Everyone who built something today - YOU ARE AN INVENTOR! Real engineers at ISRO, IIT, and Tata Motors all started exactly where you are right now! Building, testing, fixing, and trying again!

You are learning just like the best engineers in India! Every time you build and test your machine, you become a better inventor. Give yourself a BIG round of applause! You did amazing work today!





Homework Challenge

Tonight, be a detective! Find a chain reaction at home!

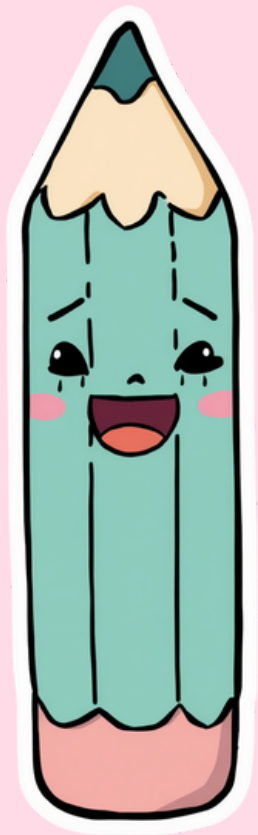
Does something fall and cause something else to happen?

Draw it in your notebook:

- What happened FIRST?
- THEN what happened?
- FINALLY what was the result?

Bring your discovery to share in next class!





See You
Next Time!

